PRECAL

Pre-Calculus

Grade 12, Semester 1 study outline

A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.

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01. The Unit Circle and Evaluating Trigonometric Functions

The Unit Circle and Evaluating Trigonometric Functions

- Topic: The Unit Circle and Evaluating Trigonometric Functions A unit circle is a circle with a radius of 1 unit.
- Therefore, for an angle in the standard position: sin = y/r = y/1 = y cos = x/r = x/1 = x tan = y/x So the coordinates of P can be written as (cos, sin).

TRIGONOMETRIC FUNCTIONS AND REFERENCE ANGLES

EVALUATING TRIGONOMETRIC FUNCTION

- Section 4.4 Trigonometric Functions of Any Angle.

PRECAL: ADDITIONAL EXAMPLES

- -- 4 of 11 -- Topic: The Unit Circle and Evaluating Trigonometric Functions 1.

The reference angle measures

The reference angle measures

The reference angle measures

The reference angle measures

The reference angle measures

The reference angle measures

PRECAL: ADDITIONAL EXAMPLES

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02. Coterminal Angles and Conversion- Degrees - Radian Measure

Coterminal Angles and Conversion- Degrees - Radian Measure

- The starting position of the ray is the initial side of the angle, and the position after rotation is the terminal side, as shown in Figure 1 An angle is in standard position, as shown in Figure 2. -- 1 of 13 -- Positive angles are generated by counterclockwise rotation, and negative angles by clockwise rotation, as...

Definition of Radian

- One radian is the measure of a central angle that intercepts an arc (s) equal in length to the radius (r) of the circle.
- See Figure 5. -- 2 of 13 -- Algebraically, this means that = s / r, where is measured in radians.

Complementary Angle

- Two angles sum up equal to /2 or 90 degrees.

Supplementary Angle

- Two angles sum up equal to or 180 degrees.

Finding Coterminal Angles

- -- 3 of 13 -- Coterminal Angles are angles who share the same initial side and terminal sides.

1. Find one positive angle and one negative angle that are coterminal with -45°

2. Find one positive angle and one negative angle that are coterminal with 395°

3. Find one positive angle and one negative angle that are coterminal with /6 radians

4. Find two positive angles that are coterminal with 11/2 radians

1. Convert 120° to radians

2. Convert 110° to radians

3. Convert - / 12 radians to degrees

4. Convert - / 9 radians to degrees
PRECAL: ADDITIONAL EXAMPLES
1. Find one positive angle and one negative angle that are coterminal with 130°
2. Find one positive angle and one negative angle that are coterminal with -60°
3. Find one positive angle and one negative angle that are coterminal with 525°
4. Find one positive angle and one negative angle that are coterminal with 645°
5. Find one positive angle and one negative angle that are coterminal with 13 / 18
6. Find one positive angle and one negative angle that are coterminal with -2 / 3
7. Find one positive angle and one negative angle that are coterminal with 35 / 12
8. Find one positive angle and one negative angle that are coterminal with 43 / 12
9. Convert 210° to radians
10. Convert 300° to radians
11. Convert 490° to radians
12. Convert 35 / 6 to degrees
13. Convert 11 / 12 to degrees
14. Convert 46 / 12 to degrees
PRECAL: ADDITIONAL EXAMPLES

Key terms

Positive angles - generated by counterclockwise rotation, and negative angles by clockwise rotation, as shown in
Angles - labeled with Greek letters like alpha, beta, and theta, as well as uppercase letters
Coterminal Angles - angles who share the same initial side and terminal sides. Finding coterminal angles is as

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03. Sum and Difference Identity(Sine Cosine)

Sum and Difference Identity(Sine Cosine)
1. Write the formula for sine
2. Substitute the given angles into the formula
3. Find the values of the special angles
4. Substitute the values into the equation
5. Simplify
1. sin(45° - 30°)
2. sin(45° + 30°)
3. sin(135° - 120°)
1. Write the formula for sine
2. Substitute the given angles into the formula
3. Find the values of the special angles
4. Substitute the values into the equation
5. Simplify
2. Find the exact value of cos(75°)
3. Find the exact value of cos (/3 - /4)
PRECAL: ADDITIONAL EXAMPLES
Note that we can express as
Note that we can express as
7. Given
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04. Sum and Difference Identity(Tangent)

Sum and Difference Identity(Tangent)

- Topic: Sum and Difference Identity (Tangent) Finding exact values for the tangent of the sum or difference of two

angles is a little more complicated, but again, it is a matter of recognizing the pattern.
- Finding the sum of two angles formula for tangent involves taking the quotient of the sum formulas for sine and cosine and simplifying.

1. Write the formula for the tangent

2. Substitute the given angles into the formula

3. Simplify

1. Find the exact value of

Solution

So we have,

2. Find the exact value of tan 75°

3. Find the exact value of tan 105°

Finding Multiple Sums and Differences of Angles

- Given that sin = 3/5, 3 is the opposite side, while 5 is the hypotenuse side.
- Given that cos = -5/13, 5 is the adjacent side while 13 is the hypotenuse side.

Similarly, we have

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1. tan 75°

2. tan 165°

We get

4. Given

Solving and using the Pythagorean triples

- - implies . - implies Both are negative since and are in the third quadrant.

Finally, we can compute and using the formula

- - -- 7 of 9 -- - Therefore, and .

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05. Double Angle Identity(Cosine)

Double Angle Identity(Cosine)

- Topic: Double Angle Identity (Cosine) Identities for angles that are twice as large as one of the common angles (double angles) are used frequently in trigonometry.
- A double-angle function is written, for example, as sin 2, cos 2, or tan 2x, where 2, 2, and 2x are the angle measures and the assumption is that you mean sin(2), cos(2), or tan(2x).
- Because tangent is equal to the ratio of sine and cosine, its identity comes from their double-angle identities.
- Because the two angles are equal, you can replace with , so cos (+) = cos cos - sin sin becomes To get the second version, use the first Pythagorean identity, sin2 + cos2 = 1.
- The sum and difference identities for sine and cosine, on the other hand, as well as the double-angle identity for sine, all involve both the sine and cosine of the angles.
- Example 1: Find cos 60° by using the functions of 30°.
- We can only use the sine and cosine functions of 30°, so we need to start with 60° = 2 x 30°.
- Use the following form of the cosine of a double angle formula, cos 2x = 1 - 2sin2 x, we have: Example 3: Find the exact value of cos 2x given cos x = -13/14 if x is in Quadrant II.
- For the remaining formulas of cosine that require the value of cos x in Quadrant III, given sin x = -3/5 setting aside the negative symbol, 3 is the opposite while 5 is the hypotenuse.
- Using the Pythagorean Theorem, x2 + 32 = 52 Simplifying it, x2 = 25 - 9, x2 = 16, therefore, x = 4 since the angle is at Quadrant III, cosine is negative, so cos x = -4/5.
- So the results are the same having 7/25 by applying the 3 formulas for cosine.
- 3.5 Double Angle Identities.

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1. Given

Find

2. Given

Find

Using the given, we know from the formula that

3. Given

Find

4. Given

Find

5. Given

Find

- 2, we can solve this item in several ways.

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06. Double Angle Identity(Sine and Tangent)

Double Angle Identity(Sine and Tangent)

Deriving the Double Angle Identities

- One of the formulas for calculating the sum of two angles is: sin (+) = sin cos + cos sin If and are both the same angle in the above formula, then sin (+) = sin cos + cos sin sin 2 = 2 sin cos This is the double angle formula for the sine function.
- We can calculate the double angle formula for tangent, using the tangent sum formula: If and are both the same angle in the above formula, then Example 1: Find sin 60° by using the functions of 30°. -- 1 of 8 -- We can only use the sine and cosine functions of 30°, so we need to start with 60° = 2 x 30°.

We can use our formula for the sine of

- So cos x = -5/13, since the angle is in Quadrant III so the value is negative. sin 2x = 2 sinx cosx = 2 (-12/13) (-5/13) = 120/169 Example 4: Find tan 60° by using the functions of 30°. -- 2 of 8 -- We can only use the tangent value of 30°, so we need to start with 60° = 2 x 30°.

We can use our formula for the tangent

- Solving for tan x = sin x / cos x, tan x = (-12/13) / (-5/13), therefore the value of tan x = 12/5. tan 2x = 2tanA / 1 - tan2A -- 3 of 8 -- = 2 (12/5) / 1 - (12/5)2 = 24/5 / 1 - 144/25 = 24/5 / (25-144)/25 = 24/5 / -119/25 = (24/5) (-25/119) = -120/119 CK-12 Foundation's .
- 3.5 Double Angle Identities.

PRECAL: ADDITIONAL EXAMPLES

1. Given

Solve

2. Given

Solve

3. Given

Solve

Now, we compute

4. Given

Solve

Now, we compute

PRECAL: ADDITIONAL EXAMPLES

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07. Half Angle Identity(Sine and Cosine)

Half Angle Identity(Sine and Cosine)

To derive the half-angle formula for cosine, we have

1. Find the exact value of sin(15°) using a half-angle formula

2. Find the exact value for sin 105° using the half-angle identity

- -- 2 of 10 -- Note that 105° is in the second quadrant, and sine functions in the second quadrant are positive.

1. Find the exact value of cos(15°) using a half-angle formula

2. Find the exact value for cos 165° using the half-angle identity

3. Use a Half Angle Identity to find the exact value of cos(/8)

Using the given information, we can draw the triangle,

- Therefore, we can calculate sin A = -8/17 and cos A = -15/17.
- Before we start, we must remember that if A is in quadrant III, then 180° < A < 270°, so 180/2 < A/2 < 270/2.
- This means that the terminal side of A/2 is in Quadrant II, since 90° < A/2 < 135°. (a).

PRECAL: ADDITIONAL EXAMPLES

1. Given

Quadrant I

Find

2. Given

Quadrant IV

Find

- Using the previous lessons, compute for . -- 7 of 10 -- The value that we should obtain is .

3. Given

Quadrant IV

Find

4. Given

Quadrant IV

Find

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08. Half Angle Identity(Tangent)

Half Angle Identity(Tangent)

- Deriving the first equation of tangent A/2: -- 1 of 9 -- Deriving the second equation of tangent A/2: Deriving the third equation of tangent A/2: -- 2 of 9 -- Use a half-angle identity for the tangent to find the exact value for tan 15°.

Using the given information, we can draw the triangle,

- Therefore, we can calculate sin A = -8/17 and cos A = -15/17.
- Before we start, we must remember that if A is in quadrant III, then 180° < A < 270°, so 180/2 < A/2 < 270/2.
- This means that the terminal side of A/2 is in Quadrant II, since 90° < A/2 < 135°.

PRECAL: ADDITIONAL EXAMPLES

1. Given

2. Given

3. Given

- -- 6 of 9 -- Find: Solution: Similar to Example No.
- 2, we need to determine the sine of .

4. Given

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09. Conic Section Classification

Conic Section Classification

Conic Section Classification

- A conic section (or simply conic) is a curve obtained as the intersection of the surface of a cone with a plane.
- The three types of conic sections are the hyperbola, the parabola, and the ellipse.
- The circle is a type of ellipse and is sometimes considered to be a fourth type of conic section.
- Its intersection with the cone is therefore a set of points equidistant from a common point (the central axis of the cone), which meets the definition of a circle.
- The definition of an ellipse includes being parallel to the base of the cone as well, so all circles are a special case of the ellipse.

- Determine the type of conic section that each general equation will produce.
- 9x2 + 9y2 - 6x + 18y + 11 = 0

A = 9, B = 0, C = 9

B2 - 4AC = (0)2 - 4 (9)(9)

- 0 - 324 = -324 < 0 Note that B = 0 and A = C.
- Thus, the conic section is a CIRCLE.
- 4x2 + 4xy + y2 + 8x + 24y + 36 = 0

A = 4, B = 4, C = 1

B2 - 4 AC = (4)2 - 4 (4)(1)

- 16 - 16 = 0 Thus, the conic section is a PARABOLA.
- 2x2 + 4y2 + 9x +24y +44 = 0

A = 2, B = 0, C = 4

B2 - 4AC = (0)2 - 4 (2)(4)

- 0 - 32 = -32 < 0 Note that A C.
- Thus, the conic section is an ELLIPSE.
- 4x2 + 6xy + 2y2 - 4x - 2y + 43 = 0

A = 4, B = 6, C = 2

B2 - 4AC = (6)2 - 4 (4)(2)

- 36 - 32 = 4 > 0 Thus, the conic section is a HYPERBOLA.
- An equation must have x^2 and/or y^2 to create a conic.
- If neither x nor y is squared, then the equation is that of a line.
- None of the variables of a conic section may be raised to any power other than one or two. -- 4 of 9 -- Certain characteristics are unique to each type of conic and hint at which of the conic sections you're graphing.
- To recognize these characteristics, the x2 term and the y2 term must be on the same side of the equal sign.
- If they are, then these characteristics are as follows: - CIRCLE.
- When x and y are both squared and the coefficients on them are the same, including the sign.
- The equation 3x2 - 9x + 2y2 + 10y - 6 = 0 is one example of an ellipse.
- The equation 4y2 - 10y - 3x2 = 12 is an example of a hyperbola.
- This time, the coefficients of x2 and y2 are different, but exactly one of them is negative and one is positive, which is a requirement for the equation to be the graph of a hyperbola. lumencandela. (n.d).

PRECAL: ADDITIONAL EXAMPLES

1. Given

- Recall: Note that the general form of conic sections is denoted by Solution: Now, we get the following: Since , and , from the conic classification, we can therefore conclude that the conic section is a CIRCLE.

2. Given

- Solution: Now, we get the following: Therefore, the conic section is a PARABOLA. -- 6 of 9 --

3. Given

Note that

4. Given

- Solution: Now, we get the following: Thus, the conic section is a HYBERBOLA.

5. Given

- 9x2 + 9y2 - 6x + 18y + 11 = 0 Answer: We can observe that both and are squared, with the same coefficients and sign.

6. Given

- Answer: The conic section is a PARABOLA, since only is squared.

7. Given

- Answer: Since and are both squared, and their coefficients are both positive but with different values, the conic section is an ELLIPSE.

8. Given

- Thus, the conic section is a HYPERBOLA.

PRECAL: ADDITIONAL EXAMPLES

Key terms

PARABOLA - formed when the plane is parallel to the surface of the cone, resulting in a U-shaped curve that lies
Here's - some more important pointers when we're given the general form of a conic section equation
Certain characteristics - unique to each type of conic and hint at which of the conic sections you're graphing. To
Since and - both squared, and their coefficients are both positive but with different values, the conic

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10. Circle(Part 1)

Circle(Part 1)

General Form of Equation

- CIRCLE is the collection of points equidistant from a fixed point.
- The fixed point is called the center.
- The distance from the center to any point on the circle is the radius of the circle, and a segment containing the center whose endpoints are both on the circle is a diameter of the circle.
- The radius, r , equals one-half the diameter, d.

The standard form of an equation of the circle,

- The general form of the equation of the circle is State the radius and center of the circle with equation 16 = (x - 2)2 + (y - 3)2 Solution: (x - h)2 + (y - k)2 = r2 is the standard equation of the circle with center at (h, k) and radius r.
- Center (0, -3) and r = 5 Find the general equation of the circle whose center is (2, 6) and whose radius is 3.
- Solution: We let (h, k) = (2, 6) and r = 3. (x - h)2 + (y - k)2 = r2 (x - 2)2 + (y - 6)2 = 32 x2 - 4x + 4 + y2 - 12y + 36 = 9 x2 + y2 - 4x - 12y + 4 + 36 - 9 = 0 -- 1 of 7 -- x2 + y2 - 4x - 12y + 31 = 0 The equation of the circle is: x2 + y2 - 4x - 12y + 31 = 0 Find the general equation of the circle whose center is...
- Solution: We let (h, k) = (9, 3) and r = 4. (x - h)2 + (y - k)2 = r2 (x - 9)2 + (y - 3)2 = 42 x2 - 18x + 81 + y2 - 6y + 9 = 16 x2 + y2 - 18x - 6y + 81 + 9 - 16 = 0 x2 + y2 - 18x - 6y + 74 = 0 The equation of the circle is: x2 + y2 - 18x - 6y + 74 = 0 Determine the general equation of the circle whose center is (3, -...

The equation of the circle is

PRECAL: ADDITIONAL EXAMPLES

General Form of Equation

1. Find the general equation of the circle

2. State the radius and center of the circle with equation

3. Find the general equation of a circle with center and radius are

4. Find the general equation of a circle

5. Find the general equation of a circle

PRECAL: ADDITIONAL EXAMPLES

General Form of Equation

Key terms

CIRCLE - collection of points equidistant from a fixed point. The fixed point is called the center. The distance

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11. Circle(Part 2)

Circle(Part 2)

Standard Form and Other Conditions

- The standard form of the equation of the circle, where the center C(h, k) and the radius r is (x-h)2 + (y-k)2 = r2 It is more convenient in the sense that we can easily identify the center and the radius of the circle.
- Another way to represent the equation of a circle, which is called the general form, is x2 + y2 + Dx + Ey + F = 0.
- We can convert the equation of a circle in general form to standard form by completing the squares, The equation may be presented as

Thus, the center and radius is equivalent to

Applying the formula,

- Note that D = 8, E = -6 and F = 0, thus The standard form is (x + 4)2+ (y - 3)2 = 25, the center is (-4, 3), and the radius is 5.

Circle Determined by Different Conditions

- Determine the equation of the circle that passes through the points P1(-1, 2), P2(0, 5) P3(2, 1).
- Solution: We start with the general equation of a circle: We will substitute the coordinates P1(-1, 2), P2(0, 5) and P3(2, 1) to establish the equation of the circle.
- 5 Substitute E = -6 in the 4th equation to obtain the value of D: Then, substitute E = -6 into the 1st equation to obtain the value of F: Lastly, we substitute , , and into the general form: Thus, the general equation of the circle is: which contains the points (-1, 2), (0, 5), and (2, 1).
- Standard Form of Circle Equation.
- Expii. https://www.expii.com/t/standard-form-of-circle-equation-5103 Mathematics Monsters, (n.d), Converting an Equation of a Circle from General to Standard Form. https://www.mathematics-monster.com/lessons/how_to_convert_an_equation_of_a_circle_from_general_form_to_standard_form.html -- 4 of 12 -- -- 5 of 12 --

PRECAL: ADDITIONAL EXAMPLES

1. Given

Rearrange the equation

Complete the square

Rearrange to standard form

Determine the center and radius

2. Given

Rearrange the equation

Complete the square

Rearrange to standard form

Determine the center and radius

3. Given

Rearrange the equation

Complete the square

Rearrange to standard form

Determine the center and radius

4. Given

Rearrange the equation

Complete the square

Rearrange to standard form

Determine the center and radius

- Determine the equation of the circle which passes through P1(-7, 6), P2(9, 6) and P3(-4, 13).
- Solution: To find the equation of the circle passing through the points , , and :

Consider the general equation given by

- 1): Similarly, subtract Eq.
- 2): Next, substitute and into Eq.1: Thus, the equation of the circle is: -- 8 of 12 --

PRECAL: ADDITIONAL EXAMPLES

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12. Parabola(Part 1)

Parabola(Part 1)

Parts and Properties of a Parabola

- General Equation of a Parabola with Vertex at (0, 0) $y^2 + Dx + F = 0$ $x^2 + Ey + F = 0$

Standard Equation of Parabola with Vertex at (0, 0)

Axis of symmetry on 0x

Axis of symmetry on 0y

- Downward opening is $x^2 = -4ay$ General Equation of a Parabola with Vertex at (h, k) $y^2 + Dx + Ey + F = 0$ $x^2 + Dx + Ey + F = 0$

Standard Equation of Parabola with Vertex at (h,k)

Axis of symmetry parallel or identical to 0x
Axis of symmetry parallel or identical to 0y
General Equation to Standard Form
Convert the general equations to standard form
Convert the general equations to standard form
Standard Equation to General Form
Convert the standard equations to general form
Convert the standard equations to general form
Parabola with Vertex at (0, 0)

- Vertex : (0, 0) Focus : (-a, 0) (-3, 0) Axis of Symmetry : x-axis Directrix : x = a x = 3 Length of L.R. : 4a 4(3) = 12 Endpoints of L.R. : E1 (-a, 2a) = (-3, 6) E2 (-a, -2a) = (-3, -6) Find the equation of the parabola with vertex at (0, 0) and focus at (0, -3/4).
- The parabola opens downward, thus the equation is x2 = -4ay Since a is the undirected distance between the vertex and the focus, it is equal to 3/4.

Figure 1 - Openings of Parabola
Figure 2 - Discussion of Parabola's Equation
Algebra and Trigonometry
PRECAL: ADDITIONAL EXAMPLES
1. Given
2. Given
3. Given
4. Given
5. Given
6. Given
7. Given
8. Given
PRECAL: ADDITIONAL EXAMPLES

Key terms

Since a - undirected distance between the vertex and the focus, it is equal to 3/4

Since the vertex - midway between the focus and directrix, its coordinate is at the origin (0, 0). Substituting in the

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13. Parabola(Part 2)

Parabola(Part 2)
Parabola with Vertex at (h, k)

- (See Figure 1 on page 3 for your guide) Give the equation of the parabola (x - 2)2 = 12 (y - 1).
- Solution: With the given standard equation of the parabola (x - 2)2 = 12 (y - 1) we will represent it in terms of (x - h)2 = 4a (y - k), we have (x - 2)2 = 4 (3)(y - 1) It follows that a = 3.
- Solution: Convert the general equation to the standard form of a parabola. -- 1 of 13 -- Note that a = 1, since (y + 3)2 = 4(1) (x + 5).
- Solution: Convert the equation to the standard form of a parabola.

Parabola Determined by Different Conditions

- Find the general equation of the parabola with Vertex (3, 6) and Focus (3, 4).
- Thus, its equation is in the form: Here, and .
- Solving for the value of a, we apply the distance formula, -- 2 of 13 -- Substituting the values of h, k, and a in the equation of the parabola, Find the general equation of the parabola with Vertex (2, 1) and directrix x = 5.
- The equation of the parabola that opens to the left is: The directrix along the x-axis of symmetry is D(h + a, k).
- Given and , substituting these values: Find the general equation of the parabola with Focus (-1, -6) and directrix y = 0.

We can now solve for the parabola,

- -- 3 of 13 -- Abramson, J., , The Parabola.

Figure 1 - Discussion of Parabola

Figure 2 - Discussion of Parabola's Equation

PRECAL: ADDITIONAL EXAMPLES

1. Given

2. Given

3. Given

4. Find the general equation of the parabola that satisfies the given condition

Vertex (4, 8), Focus (4, 6)

- Solution: Since the x-coordinates of both the vertex and focus are the same, the orientation of the parabola is vertical, and it opens downward because the focus is below the vertex.

We have Vertex (4, 8), Focus (4, 6), then

Thus, we get

Therefore, the general equation of the parabola is

5. Find the general equation of the parabola that satisfies the given condition

Thus, we get

Therefore, the general equation of the parabola is

6. Find the general equation of the parabola which satisfies the given condition

- Since the directrix is horizontal, i.e. , the parabola's axis of symmetry is a vertical line passing through the focus.
- Note that the vertex is the midpoint between the focus and the directrix along the axis of symmetry so the y-coordinate of the vertex is the midpoint between the y-coordinates of the focus and the directrix, that is .
- The x-coordinate of the vertex is the same as the x-coordinate of the focus because the parabola is symmetric about this vertical line.

Now, calculate

Now, we have

Therefore, general equation of the parabola is

PRECAL: ADDITIONAL EXAMPLES

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14. Ellipse(Part 1)

Ellipse(Part 1)

Important Lengths and Distance Involve in Ellipse

Properties of Ellipse

General Equation to Standard Form

- 9x2 + 8y2 = 288 Solution: -- 1 of 11 -- Convert the general equations to standard form.
- 25x2 + 49y2 = 1,225 Solution: Convert the general equations to standard form.
- 9x2 + 25y2 = 225 Solution: Convert the general equations to standard form.
- 4x2 + 9y2 = 36 Solution: Convert the general equations to standard form.
- 3x2 + 4y2 + 24x - 16y + 52 = 0 Solution: Regroup the terms. (3x2 + 24x) + (4y2 - 16y) = -52 Apply common factor.
- 3(x2 + 8x) + 4(y2 - 4y) = -52 Apply completing the square.
- 3(x2 + 8x + 16) + 4(y2 - 4y + 4) = -52 + 3(16) + 4(4) Simplify the right side. -- 2 of 11 -- 3(x2 + 8x + 16) + 4(y2 - 4y + 4) = -52 + 48 + 16 3(x2 + 8x + 16) + 4(y2 - 4y + 4) = 12 Factoring completely.
- 3(x + 4)2 + 4(y - 2)2 = 12 Divide both sides by 12.
- 25x2 + 16y2 + 150x - 128y - 1119 = 0 Solution: Regroup the terms. (25x2 + 150x) + (16y2 - 128y) = 1,119 Apply common factor.
- 25(x2 + 6x) + 16(y2 - 8y) = 1,119 Apply completing the square.
- 25(x2 + 6x + 9) + 16(y2 - 8y + 16) = 1,119 + 25(9) + 16(16) Simplify the right side.
- 25(x + 3)2 + 16(y - 4)2 = 1,600 Divide both sides by 1600.

Standard Equation to General Form

- Lengths, Distances, Properties of an Ellipse -- 5 of 11 -- Ikelyn (n.d), Transform a general equation of an ellipse to the standard form by completing the square.

PRECAL: ADDITIONAL EXAMPLES

1. Given

2. Given

3. Given
4. Given
5. Given
PRECAL: ADDITIONAL EXAMPLES

15. Ellipse(Part 2)

Ellipse(Part 2)

Ellipse with Center at the Origin

- (See Figure 1 as your guide) Discuss: 4x2 + 9y2 = 144 Solution: Since the denominator of x2 is greater than the denominator of y2, the major axis is the x-axis.

Ellipse with Center at (h, k)

- Solution: 36x2 + 100y2 - 72x + 200y -3,464 = 0 (36x2 - 72x) + (100y2 + 200y) = 3,464 36 (x2 - 2x + 1) + 100 (y2 + 2y + 1) = 3,464 + 36 + 100 36 (x - 1)2 + 100 (y + 1)2 = 3,600 Since the denominator of (x - 1)2 is greater than the denominator of (y + 1)2, the major axis is the x-axis.
- Solution: 25x2 + 9y2 - 150x - 36y + 36 = 0 (25x2 - 150x) + (9y2 - 36y) = - 36 25 (x2 - 6x + 9) + 9 (y2 - 4y + 4) = - 36 + 225 + 36 25 (x - 3)2 + 9 (y - 2)2 = 225 Since the denominator of (y - 2)2 is greater than the denominator of (x - 3)2, the major axis is the y-axis.
- Solution: 4x2 + y2 - 8x - 4y - 28 = 0 (4x2 - 8x) + (y2 - 4y) = 28 4 (x2 - 2x + 1) + (y2 - 4y + 4) = 28 + 4 + 4 4 (x - 1)2 + (y - 2)2 = 36 Since the denominator of (y - 2)2 is greater than the denominator of (x - 1)2, the major axis is the y-axis.
- Solution: 9x2 + 16y2 - 54x - 64y + 1 = 0 (9x2 - 54x) + (16y2 - 64y) = -1 9 (x2 - 6x + 9) + 16 (y2 - 4y + 4) = -1 + 81 + 64 9 (x - 3)2 + 16 (y - 2)2 = 144 Since the denominator of (x - 3)2 is greater than the denominator of (y + 2)2, the major axis is the x-axis.

Figure 1 - Vertex at Origin

Figure 2 - Vertex at (h, k)

PRECAL: ADDITIONAL EXAMPLES

1. Given

Thus,

We get

2. Given

Thus,

We get

- -- 13 of 20 -- Center: Vertices : Co-Vertices : Foci : Endpoints of Latera Recta: Directrices : Eccentricity : Length of Latus Rectum : Length of Major Axis : Length of Minor Axis :

3. Given

Then,

Hence,

We get

4. Given

Then,

Hence,

Thus, we have

PRECAL: ADDITIONAL EXAMPLES

16. Hyperbola(Part 1)

Hyperbola(Part 1)

- Topic: Hyperbola - General Form and Standard Form Equation A hyperbola is the set of all points in a plane such that the absolute difference of the distances from any point on the curve to two fixed points (called the foci) is a constant, equal to 2a.

Important Lengths and Distances Involved in a Hyperbola

General Equation to Standard Form

- 3x2 - 2y2 = 12 Solution: Convert the general equations to standard form.
- 9y2 - 16x2 = 144 Solution: Convert the general equations to standard form. x2 - 9y2 = 9 Solution: Convert the general equations to standard form.
- 9x2 - 4y2 + 18x - 16y - 43 = 0 Solution: Regroup the terms. (9x2 + 18x) + (-4y2 - 16y) = 43 Apply common factor.
- 9(x2 + 2x) - 4(y2 + 4y) = 43 Apply completing the square.
- 9(x2 + 2x + 1) - 4(y2 + 4y + 4) = 43 + 9 + (4)(-4) Simplify the right side.
- 9(x2 + 2x + 1) - 4(y2 + 4y + 4) = 43 + 9 - 16 9(x2 + 2x + 1) - 4(y2 + 4y + 4) = 36 Factoring completely.
- 9(x + 1)2 - 4(y + 2)2 = 36 -- 3 of 12 -- Divide both sides by 36.

Standard Equation to General Form

- General Form of the Equation.

Figure 1 - Parts of a Hyperbola

PRECAL: ADDITIONAL EXAMPLES

1. Given

2. Given

3. Given

4. Given

5. Given

PRECAL: ADDITIONAL EXAMPLES

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17. Hyperbola (Part 2)

Hyperbola (Part 2)

Hyperbola at Center (0, 0)

Hyperbola with Center at (h, k)

Hyperbolas

Figure 1 - Center at (0, 0)

Figure 2 - Center at (h, k)

PRECAL: ADDITIONAL EXAMPLES

1. Given

Thus,

- -- 10 of 16 -- Center: Vertices : Foci : Endpoints of Conjugate Axis : Endpoints of Latera Recta: Asymptotes: Eccentricity : Length of Transverse Axis : Length of Conjugate Axis : Focal Length: Length of Latus Rectum:

2. Given

Thus,

Hence, the endpoints are

3. Given

Thus,

4. Given

Thus,

PRECAL: ADDITIONAL EXAMPLES

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18. Systems of Nonlinear Equations

Systems of Nonlinear Equations

- Topic: System of Nonlinear Equations A system of nonlinear equations is a system of two or more equations in two or more variables containing at least one equation that is not linear.
- Any equation that cannot be written in this form is nonlinear.
- The substitution method we used for linear systems is the same method we will use for nonlinear systems.

Intersection of a Parabola and a Line

1. Solve the linear equation for one of the variables

2. Substitute the expression obtained in step one into the parabola equation

3. Solve for the remaining variable

4. Check your solutions in both equations

Intersection of a Circle and a Line

1. Solve the linear equation for one of the variables

2. Substitute the expression obtained in step one into the equation for the circle

3. Solve for the remaining variable

4. Check your solutions in both equations

Points of Intersection of a Circle and an Ellipse

PRECAL: ADDITIONAL EXAMPLES

1. Sketch the graphs of the equations x2 - y + 1 = 0 and y - 2x = 0 and find the coordinates

2. Sketch the graphs of the equations x2 - y = 0 and x2 + y - 8 = 0 and find the coordinates

Using elimination,

4. Solve the following system y = x2 + 2x + 1 and y = 2x2 + 3x + 2, and sketch the graph

- Note that there is a negative number inside the square root (-3) and it can't be graphed.

Note that

If , we have If , we have

- -- 9 of 15 -- Thus, the points of intersection are .

Figure 1 - Involving Parabola and a Line

Figure 2 - Involving Circle and a Line

Figure 3 - Involving Circle and Ellipse

PRECAL: ADDITIONAL EXAMPLES